

# Nils Duran

Machine Learning (ML) Engineer | MSc Data Science Student

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## Experience

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>Sparsity Technologies</b> , Data Engineering Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Barcelona, Catalonia<br>Oct 2025 – Jan 2026<br>4 months |
| <ul style="list-style-type: none"><li>Engineered automated data collection and ingestion pipelines for high-performance graph database solutions.</li><li>Executed complex graph queries and data transformation workflows for competitor intelligence and marketing analytics.</li></ul>                                                                                                                                                                                                                                                                                  |                                                         |
| <b>Werfen</b> , Software Tester Intern — Beacon Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Barcelona, Catalonia<br>June 2026 – present<br>4 months |
| <ul style="list-style-type: none"><li>Engineered full-stack automated testing suites in Java for in-vitro diagnostic (IVD) laboratory software on the Beacon Project.</li><li>Implemented comprehensive API tests and FormUI end-to-end automation, verifying data consistency across clinical workflows.</li><li>Managed requirements traceability, test planning, and regulatory compliance validation using Jama Connect.</li><li>Built custom utility scripts and automation tooling that eliminated repetitive manual QA tasks and accelerated test cycles.</li></ul> |                                                         |

## Education

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| <b>MSc</b> <b>Universitat Politècnica de Catalunya (UPC)</b> , Data Science                                                          | Barcelona, Catalonia<br>Sept 2026 – present   |
| <ul style="list-style-type: none"><li>Distributed data architectures, statistical modeling, and scalable machine learning.</li></ul> |                                               |
| <b>BSc</b> <b>Universitat Politècnica de Catalunya (UPC)</b> , Artificial Intelligence                                               | Barcelona, Catalonia<br>Sept 2022 – June 2026 |
| <ul style="list-style-type: none"><li>Thesis on EEG cognitive decoding using Spatio-Temporal GNNs, Transformers, and LoRA.</li></ul> |                                               |
| <b>Exchange</b> <b>Karlsruhe Institute of Technology (KIT)</b> , Computer Science                                                    | Karlsruhe, Germany<br>Oct 2024 – Mar 2025     |
| <ul style="list-style-type: none"><li>Advanced distributed algorithms and high-performance systems. German C1 certified.</li></ul>   |                                               |

## Projects

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| <b>Sinergia — Multi-Agent LLMs (Telefónica Colab)</b>                                                                                                                                                                                                                                                                                                      | May 2025  |
| <ul style="list-style-type: none"><li>Designed multi-expert LangGraph agent architecture with LoRA fine-tuning, boosting MedQA accuracy by +3% absolute over Gemini Flash.</li><li>Evaluated diversity dissimilarity matrices and Elo rankings; developed interactive Streamlit app (<a href="#">Public Repo</a>).</li></ul>                               |           |
| <b>HPC Vision Transformers on MareNostrum 5 (BSC)</b>                                                                                                                                                                                                                                                                                                      | June 2024 |
| <ul style="list-style-type: none"><li>Trained Vision Transformers on the 200 GB+ MAME dataset using the MareNostrum 5 supercomputer (top 10 public supercomputer worldwide).</li><li>Orchestrated distributed multi-GPU training with Slurm and PyTorch, achieving a 0.75 F1-score across hundreds of fine-grained categories.</li></ul>                   |           |
| <b>Vegan Tools — Production AI Web Platform</b>                                                                                                                                                                                                                                                                                                            | Sept 2026 |
| <ul style="list-style-type: none"><li>Architected and deployed full-stack platform featuring Ingredient Checker, Menu OCR, and Recipe Vega-nizer (<a href="#">Live at vegantools.org</a>).</li><li>Engineered end-to-end multi-modal AI API integration, client state management, and custom cloud infrastructure (<a href="#">Public Repo</a>).</li></ul> |           |

## Skills

**ML & Deep Learning Frameworks:** PyTorch, Scikit-learn, LangGraph, LoRA / PEFT, Hugging Face Transformers, Gymnasium (RL), XGBoost

**Engineering & Deployment:** Python, Slurm (MareNostrum 5 HPC), Docker, Git/GitHub, REST APIs, Streamlit, Linux, Bash

**Data & Backend:** PostgreSQL, SQL, Graph Databases (Sparsity), Apache Spark, Vector Search, Automated Testing (Java)

**Languages:** Catalan (Native), Spanish (Bilingual), English (C2 Proficient), German (C1 Advanced)